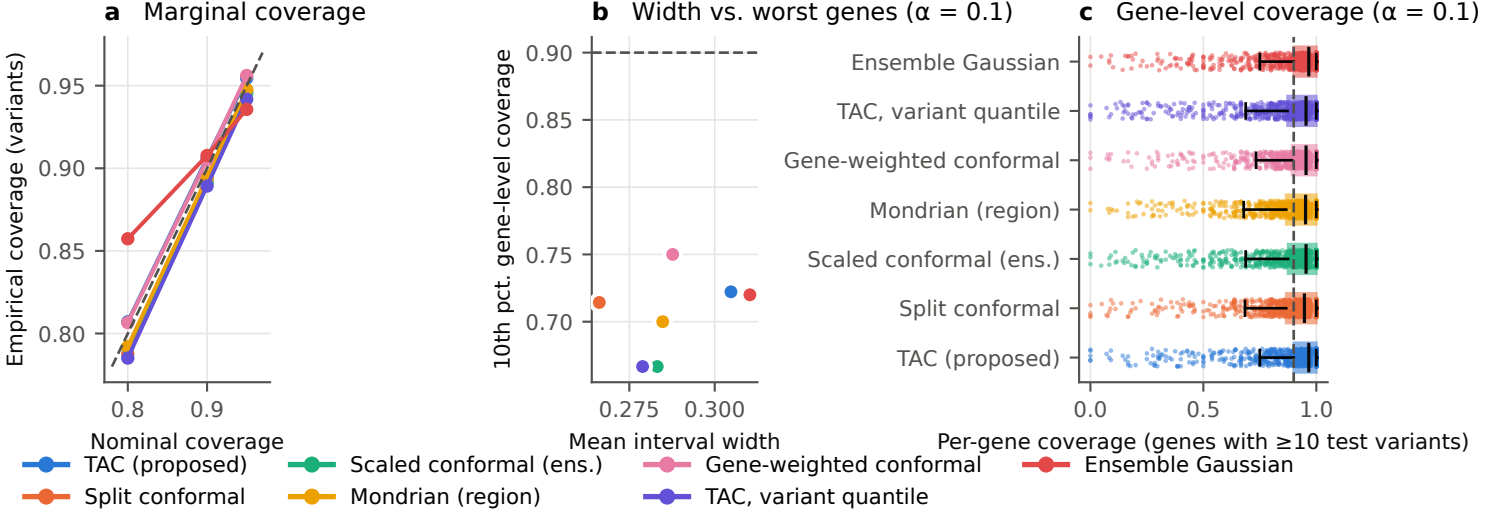

a Marginal coverage
Empirical coverage (variants)
0.95
0.90
0.85
0.80
0.8
0.9
Nominal coverage
b Width vs. worst genes (α = 0.1)
10th pct. gene-level coverage
0.90
0.85
0.80
0.75
0.70
0.275
0.300
Mean interval width
c Gene-level coverage (α = 0.1)
Ensemble Gaussian
TAC, variant quantile
Gene-weighted conformal
Mondrian (region)
Scaled conformal (ens.)
Split conformal
TAC (proposed)
0.0
0.5
1.0
Per-gene coverage (genes with ≥10 test variants)
TAC (proposed)
Split conformal
Scaled conformal (ens.)
Mondrian (region)
Gene-weighted conformal
TAC, variant quantile
Ensemble Gaussian